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Note About Implicit Function Theorem

The Implicit Function Theorem, whose proof relies on that of the Inverse Function Theorem (by converting the given map F to a map between same dimensional spaces and then inverting), gives us a ball around a solution point. So consider:

$$\exists (x_0, y_0) \in \mathrm{Dom}(F) : F(x_0, y_0) = 0$$

Then we see that $\exists \delta > 0, \exists y(x), \forall x \in B(x_0, \delta) : F(x, y(x)) = 0$. We now discuss the idea of globally extending this solution to (possibly) all of the domain. The Implicit Function Theorem requires us to prove an invertible Jacobian y-minor at the point (x_0, y_0). If this is true for all $(x, y) \in \mathbb{R}^{m+n}$, then we can explore global extensions (note that while the continuity of the Jacobian is sufficient, the existence of the Jacobian at each point is also sufficient).

Now assume there exists some $\delta > 0$ such that we have an implicit solution in the ball $B(x_0, \delta)$. Then take some $0 < \delta_1 < \delta$, then we have that $F(x_0 + \delta, y)$